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Intensified CCD Imaging and Spectroscopy Unravel the Mysteries of Carbon Nanotube Formation

Carbon nanotubes are hollow, cylindrical tubes formed by single layers of carbon atoms. They can be one atom layer thick (single-wall nanotube, or SWNT) or multiple layers thick (MWNT) with additional graphene layers forming concentrically aligned cylinders (see Figure 1).

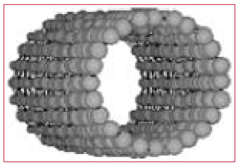

Figure 1: An SWNT showing conjoined hexagonal rings of carbon (as in a graphite sheet) rolled into a tube. The tubes can be produced open-ended or end-capped by manipulation of physical and chemical conditions.

SWNTs are formed by laser vaporization, dc-arc vaporization, chemical vapor deposition, or gas disproportionation in the presence of metal catalyst nanoparticles in background gases. Recently, SWNTs were formed by annealing C60 and Ni films in vacuum. SWNTs are another allotrope of solid carbon, joining the family of graphite, diamond, and solid fullerenes. They are the latest discovery in the field of carbon nanomolecules that began in the 1980s with the discovery of "Buckyballs", symmetrical carbon-atom spheres (named Buckminsterfullerenes) that resemble soccer balls. Like other carbon allotropes, the distinct characteristics of SWNTs are conveyed by the propensity of carbon atoms to bond to one another and form the ubiquitous planar hexagonal rings, as in graphite or the benzene molecule.

Potential Applications for Nanotubes

SWNTs have extremely high length-to-diameter ratios. Their diameters are approximately 1 to 2 nm, with known lengths up to 1 mm. Although their diameter is only 1/10,000 of a human hair, SWNTs have been estimated to be 10 times lighter and 1000 times stronger than a steel rod of the same size. They can also have metallic characteristics with unique electrical properties.

SWNTs can associate into linear bundles or ropes, which suggests that they might serve some revolutionary structural and electronic uses, including:

- Nanoelectronics — carbon-based nanotransistors; replacement for silicon as components become nanosized; nanogenerators and machines
- Nanoshielding — protection of enclosed materials (e.g., metals and drug molecules) from outside influences such as oxidation
- Hydrogen containers — safe storage of hydrogen for use as a fuel
- Lubrication — solid lubricants for use in place of petroleum products
- Structural materials — ultrastrong, ultralight building materials
- Tethers — super tethers for energy or momentum transfer in space or between earth and space

Carbon Nanotube Formation

Investigation of Nanotube Formation

For nanotubes to be fabricated into useful products, their formation will have to be clearly understood and the dynamics will have to be modifiable and controllable. However, understanding of nanotube synthesis is in its infancy and controlled nanotube production has yet to be realized. Drs. David Geohegan (www.ornl.gov/~odg) and Alexander Puretzky and their colleagues at Oak Ridge National Laboratory (ORNL) use laser ablation, as well as laser-induced luminescence (LIL), incandescence, and scattering measurements, to study the dynamics of nanotube formation. In their experiments, a small amount of material (~10^{16} carbon atoms plus ~10^{14} metal catalyst atoms [Ni and Co] from a 1-inch composite target) is vaporized by a laser pulse (Nd:YAG, 8-ns FWHM) in inert gas in a small tube furnace (see Figure 2). Under the proper conditions, a portion of the vaporized material self-assembles to form SWNTs. Experimental conditions are varied, and the resulting images and spectroscopic measurements have provided some of the most detailed data ever on the formation dynamics of SWNTs. The ORNL investigators use a Princeton Instruments PI-MAX™ intensified CCD (ICCD) camera system for imaging and spectroscopy. The high sensitivity and intensifier's fast gating make this the ideal system for their experiments, as does the dual imaging and spectroscopy capability of the PI-MAX camera.

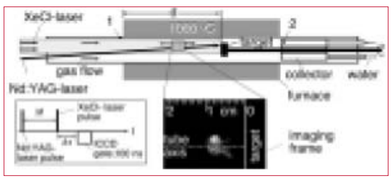

Figure 2: A schematic of the ORNL laser-vaporization setup. A 2-inch-diameter quartz tube and 1000°C furnace were used. Beam geometries and imaging area are shown. The black dots and numbers indicate the collection points of ablated material: 1 = upstream, 2 = collector. The C/Ni/Co target was placed at different distances, d, from the front of the furnace.

Dynamics of SWNT Formation

In the ORNL experiments, nanotubes were synthesized under variable laser repetition rates, flow conditions, target positions, and numbers of shots on the target. Resulting product deposits were analyzed by brightfield TEM (transmission electron microscopy) and correlated with the transport dynamics observed during the run by either time-resolved imaging or spectroscopy. Figure 3 shows typical bundles (~10 µm in length) of SWNTs and residue material from the collector (formed from only a single laser shot) when the laser was placed 21 cm from the target and operated at 0.016 Hz. To investigate nanotube growth, the C/Ni/Co plume was examined at different times after laser vaporization ranging from 20 ns to 3 s.

Figure 4 shows ICCD images of the nascent plasma emission from the plume at times up to 200 µs both at 1000°C and room temperature. The images demonstrate oscillations and self-focusing effects during the early plume dynamics at each of the temperatures.

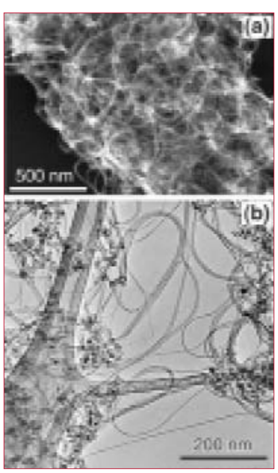

Figure 3. A TEM image of raw soot from the collector (point #2) with d = 21 cm. There is a high proportion (~90%) of SWNT bundles, along with metal nanoparticles (black dots) and some amorphous carbon.

Carbon Nanotube Formation

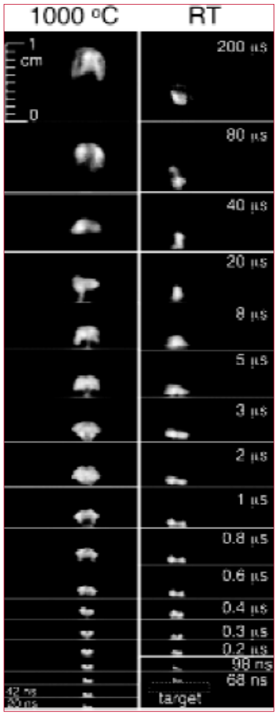

Figure 4. ICCD images of plumes at 1000°C and room temperature at various times after laser vaporization.

Laser-induced incandescence images also recorded plume dynamics during SWNT synthesis at times after ablation (see Figure 5). At 1000°C, the leading edge of the plume propagates with velocities ranging from 10^3 cm/s (early on) to 6 cm/s (at later times). After 2 s, the plume stops moving upstream and the plane of the vortex ring tilts toward the tube axis. The plume is then dragged by the gas flow back to the collector at a flow velocity of 0.6 cm/s, where nanotubes and residual material deposit on the cool collector surface by thermophoresis. At room temperature, the plume propagates slower in the axial direction and the motion of material in the plume is extremely turbulent.

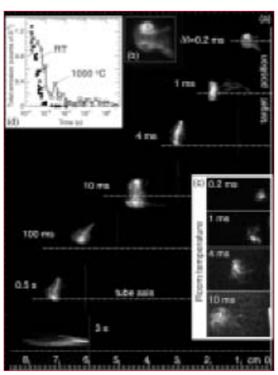

Figure 5. (a) = LIL images of plume dynamics. The Nd:YAG laser vaporized the target (right edge of figure) at 1000°C in 500-Torr argon (flowing to the right at 100 sccm). Each image is a different ablation event (100-ns gate width, opened Δt = 100 ns after the XeCl-laser pulse). (b) = Enlarged view (2x) of the plume at 0.2 ms showing vortex ring. (c) = LIL images at room temperature in 500-Torr argon. (d) = Integrated total emission from LIL images acquired at the indicated times at 1000°C and room temperature.

Rayleigh-scattering images were used to estimate the onset of plume condensation into nanoparticles after KrF-laser ablation at room temperature. These measurements yielded an estimate of 150 µs for the onset time. The images also revealed that despite the turbulent expansion of the plume, the ablated material remained confined to a relatively small volume within thin sheets of multiple vortices.

OES (optical emission spectroscopy) and LIL spectra were obtained at 1000°C and room temperature. Figure 6 shows summary spectra from room-temperature experiments. These data indicated that at early times in the plume expansion close to the target while the plasma is very hot, the plume species are primarily electronically excited. This emission from excited states dominates any laser-induced luminescence from the ground states. As the plasma expands, cools, and recombines, the ground states become populated, and LIL-emission emerges and competes with the nascent plasma emission. Finally, the nascent plasma emission completely disappears, leaving only LIL from ground states. Disappearance of plasma emission presumably signals the onset of nanoparticle formation.

Carbon Nanotube Formation

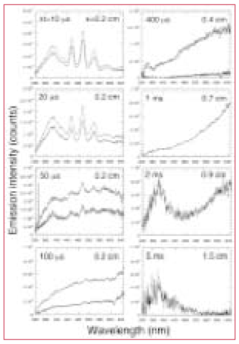

Figure 6. Plasma emission (light curves) and LIL (black curves) spectra measured at room temperature at different times (Δt) after the laser pulse and at different distances (x) from the target.

Once sharp spectral features were identified in a set of spectroscopic data, optical filters were used with ICCD imaging to selectively picture different constituents of the plume (spectroscopic imaging). For example, the 320 to 380-nm spectral region was imaged at 1000°C to observe the groundstate atomic Co in the plume. Figure 7 shows results of these experiments. The data were consistent with sequential condensation of carbon and cobalt into clusters. This clustering presumably initiates nanotube formation; nanotubes subsequently grow in a vortex ring from a feedstock of mixed nanoparticles during seconds of time.

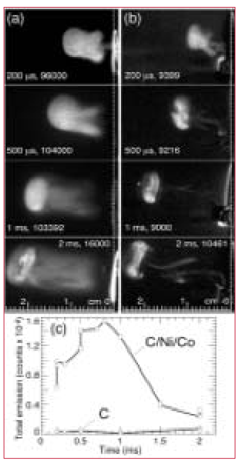

Figure 7. Selective imaging in the 320 to 380-nm spectral range at 1000°C in 500-Torr argon (100-ns gate width, Δt = 0, peak image intensities listed). (a) = Ground-state Co in the plume during SWNT synthesis. (b) = Carbon species in the same region. (c) = Total emission intensities from sets of images as in (a) and (b) comparing the Co emission temporal history with that of carbon species in the same spectral region.

To check this conclusion and estimate SWNT growth rate, experiments were performed with the target placed closer to the front furnace edge (d = 12.5 cm). The plume motion seen in Figure 8 is similar to that observed previously, though at times >100 ms the plume propagation changed such that the plane of the ring vortex tilted relative to the tube axis. The ring also elongated along this axis because of the flow gradients in the quartz tube. At 0.5 to 0.7 s, the plume exited the furnace in this orientation to deposit on the upper surface of the tube. The inset in Figure 8 is a TEM image of this deposit showing a collection of aggregated carbon and metal catalyst nanoparticles and thin SWNT bundles only ~100 nm in length. Therefore, the time spent by the plume in the hot zone (~0.5 s) was not sufficient to convert all of the carbon into nanotubes. Based on results like these, the ORNL scientists estimated that the average SWNT growth rate at 1000°C was ~1 to 5 μm/s.

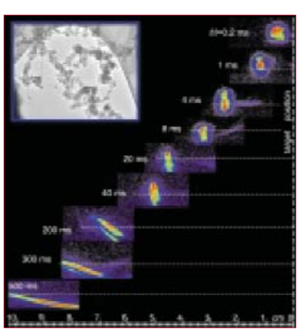

Figure 8. (a) = LIL images of the C/Ni/Co plume during synthesis of SWNTs with controlled growth times of ~0.5 s. (b) = TEM of deposit collected from point #1 showing short (~100 nm) SWNTs in the early stages of growth.

Carbon Nanotube Formation

Conclusions

The elegant work of Drs. Geohegan and Puretzky and their associates has yielded the following important information about the dynamics of SWNT formation:

- Long SWNTs (i.e., ~10 µm) can be created from a very small amount of material with a single laser shot, at repetition rates as low as 0.016 Hz.
- The average growth rate is ~1 to 5 µm/s at 1000°C with laser vaporization.
- The ablation plume initially consists of atomic and molecular species, with no evidence of hot, molten particulates.
- Condensation of carbon occurs within 0.2 ms of ablation, with Co condensing much later (i.e., 1.5 to 2 ms).
- After this time, nanotubes grow within a plume vortex ring in a ~1-cm^3 volume during seconds after laser ablation.
- Nanotubes grow during such long times (for all times after a few ms) from the condensed phase conversion of the carbon clusters and nanoparticles by the metal catalyst nanoparticles.

The latter result was confirmed recent ex situ annealing experiments deposits collected from the laser-oven after limiting nanotube oven with in situ diagnostics.

PI-MAX ICCD Camera Systems

State-of-the-art Princeton Instruments PI-MAX ICCD cameras offer better temporal resolution (gate widths <2 ns), superior background discrimination (MCP Bracket Pulsing™ with on/off ratios >107:1 in the UV), and greater sensitivity (QE >40%) than earlier-generation ICCDs. PI-MAX systems employ an integrated Programmable Timing Generator™ that allows the user to set all gating parameters under GUIsoftware control. Optimized PI-MAX$_{MG}$ ICCD systems utilize novel technology to achieve gating speeds of less than 9 ns, while preserving the highest possible QE of the photocathode. The WinSpec and WinView software packages provide stand-alone 32-bit spectral and image acquisition, display, processing, and archiving functions. Each software package is customized to provide maximum ease and versatility for spectroscopic and imaging applications.

References:

1. A.A. Puretzky, D.B. Geohegan, X. Fan, and S.J. Pennycock. In situ imaging and spectroscopy of single-wall carbon nanotube synthesis by laser vaporization. Appl. Phys. Lett. 76: 182-184, 2000.

2. A.A. Puretzky, D.B. Geohegan, X. Fan, and S.J. Pennycock. Dynamics of single-wall carbon nanotube synthesis by laser vaporization. Appl. Phys. A, 70: 153-160, 2000.

Figure 1 courtesy of Dr. Riichiro Saito, University of Electrocommunications (Tokyo, Japan) in collaboration with Prof. Mildred S. Dresselhaus, MIT (Cambridge, MA).

Figures 2 - 8 courtesy of Drs. David Geohegan and Alexander Puretzky, Oak Ridge National Laboratory (Oak Ridge, TN).